SMART INDIA HACKATHON 2025

TITLE PAGE

Problem Statement ID - SIH25220

Problem Statement Title - Digital Contact Tracing and Screening Tool for MDR Pathogens in Hospitals

Theme - MedTech / BioTech / HealthTech

PS Category - Software

Team ID - 92543

Team Name - Amrit Sparsh

Amrit Sparsh

IDEA TITLE

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About Amrit Sparsh →

A digital solution to track and control MDR pathogens in hospitals with AI and real-time alerts."

Problem Snapshot

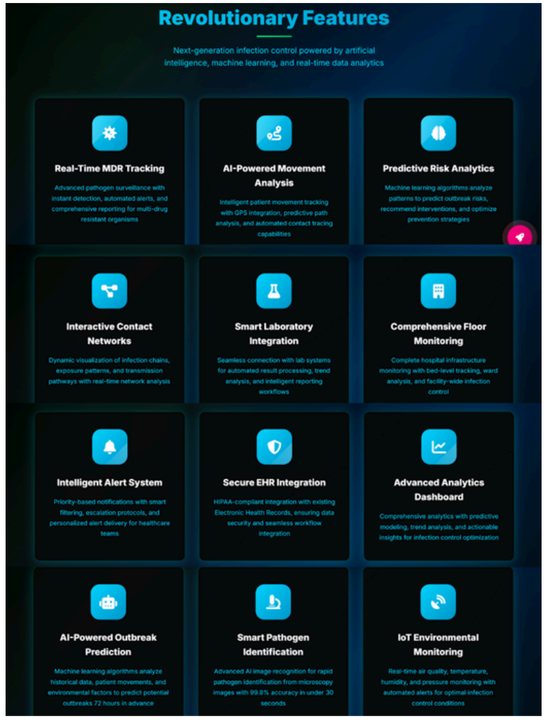
- Most infection tracking is manual , slow & error-prone
- Patient contact & movement data not recorded digitally
- Delayed detection of MDR outbreaks
- No real-time alerts for hospital staff
- Limited preventive tools

Proposed Solution and How It Solves Problems

- Centralized digital infection-control platform
- Real-time patient tracking & MDR exposure alerts
- AI-powered outbreak prediction
- Seamless EHR & lab integration
- Replaces manual tracking with instant alerts
- Enables early risk detection & preventive action
- Reduces costs and patient safety risks

Innovation & Uniqueness

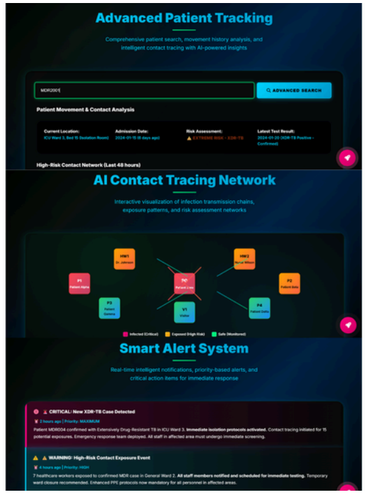
- First digital MDR tracing system in India
- Real-time exposure chain detection
- AI-powered risk alerts & outbreak prediction
- Automated containment suggestions
- Secure, scalable, hospital-ready

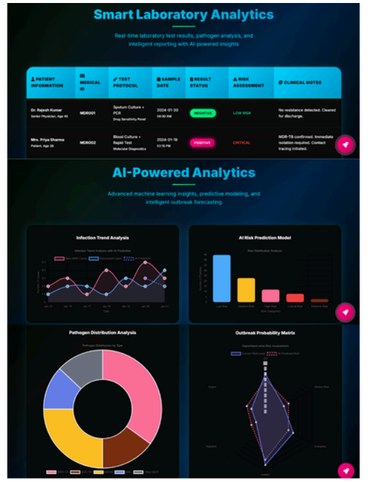






Amrit Sparsh UI Preview





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TECHNICAL APPROACH

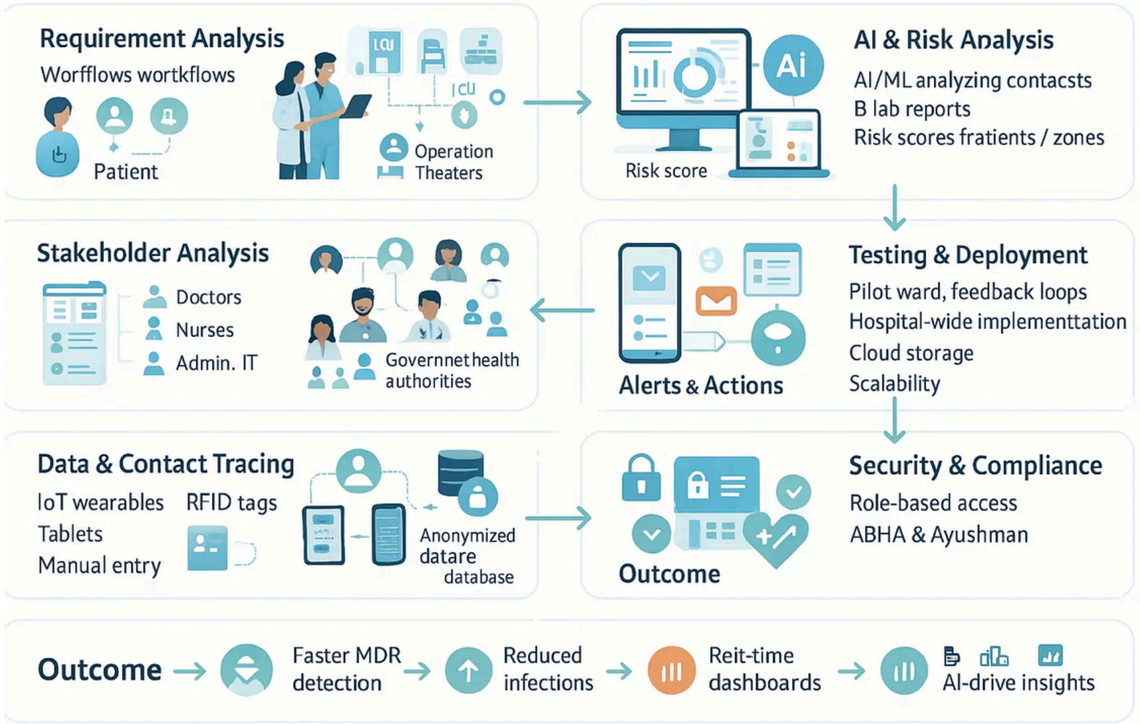
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Innovative Tech Stack

- **Frontend : HTML, CSS, JS, React , Flutter (mobile)**
- **Backend : Node.js / Python**
- **Database : PostgreSQL / MongoDB**
- **Cloud Services : AWS / Firebase**
- **AI/ML : TensorFlow / PyTorch – MDR risk prediction**
- **Security: AES-256 encryption, role-based access**
- **Integration: Hospital EHR, ABHA & Ayushman Bharat IDs**
- **Extras : Real-time alerts, IoT wearables**

Methodology and Process for implementation



"Patient → Tracking → AI Risk → Alerts → Doctors → Records → ABHA / Hospital EHR"

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FEASIBILITY AND VIABILITY

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Feasibility Analysis

Technology :- Open-source (Flask, Node.js, PostgreSQL, React), AI/ML, cloud → scalable & cost-effective

Operational :- Cost ₹8–12L/hospital, reduces paperwork, saves staff time

Market :- Solves hospital MDR gaps, supports Govt. digital health initiatives, validated in pilots



Challenges & Risks

- Data Privacy & Security: Protecting sensitive patient and hospital data
- Adoption Resistance: Staff may hesitate to switch from manual workflows
- Integration: Smooth connection with EHR, lab systems, ABHA required
- Alert Reliability: Network or IoT failures may delay notifications
- Funding & Scaling: Nationwide deployment needs strong support

Strategies To Overcome

- **Security & Privacy :-** End-to-end encryption, role-based access, compliance audits
- **Adoption :-** User-friendly dashboards, training, and step-by-step onboarding
- **Integration :-** Start with pilots → scale to ABHA/Govt. programs
- **Alert Reliability :-** Offline-first alerts, local backup notifications, IoT redundancy
- **Funding & Scaling :-** Leverage Govt. schemes, incubators, and NGO/CSR partnerships

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IMPACT AND BENEFITS

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Impact on Target Audience

Doctors & Staff
- Real-time MDR alerts
- Centralized patient & infection data
- Less paperwork, faster response

Infection-Control Teams
- Quick outbreak detection
- Visual dashboards & exposure chains
- Data-driven containment

Patients & Visitors
- Safer hospital environment
- Reduced infection risk
- Faster preventive care

Government / Health Authorities
- Accurate MDR monitoring
- Supports digital health initiatives



Benefits

Social
- Saves lives via early alerts
- Preventive healthcare
- Inclusive & accessible

Economic
- Cuts admin & paperwork costs
- Affordable & scalable
- Enables Govt.–startup partnerships

Environmental
- Paperless records
- Less storage & logistics waste
- Supports sustainable hospitals

RESEARCH AND REFERENCES

Real Life Research For Amrit Sparsh

- **JP Nadda & Bill Gates** – Discussed India's healthcare progress and the role of AI in enhancing healthcare services.

https://timesofindia.indiatimes.com/india/bill-gates-meets-jp-nadda-in-parliament-discusses-healthcare-progress-and-future-collaboration/articleshow/119203513.cms

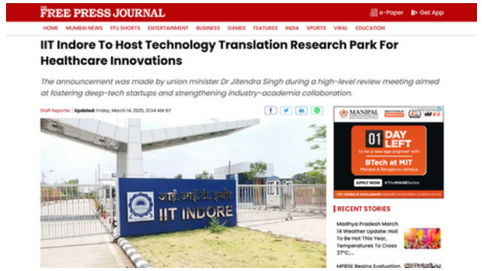
National Emphasis on AI In Healthcare



- **IIT Indore** – Technology Translation Research Park (TTRP) – A healthcare-focused park aimed at promoting innovations and deep-tech startups.

https://www.freepressjournal.in/indore/iit-indore-to-host-technology-translation-research-park-for-healthcare-innovations?utm_source=chatgpt.com

IIT Indore : Future Healthcare Hub



- **DISHA 2.0** – Initiative by IITI DRISHTI CPS Foundation to support commercialization of faculty/student-led digital healthcare innovations and DPIIT-registered startups.

https://drishticps.iiti.ac.in/disha-2-0/

- **Government & IIT Competitions –** Organized to promote innovations in Healthcare management systems.



- **Field Visits –** Conducted at Choithram Hospital, Unique Hospital, and Aurobindo Hospital to study infection control workflows, patient data management, MDR tracking practices, and integration with existing hospital systems for solution alignment.





Link To Visit Amrit Sparsh FrontEnd https://mdramritsparsh.netlify.app/